R-Slip Robot Experiment - Complete Setup Guide

🚀 Quick Start (3 Steps)

1. **Run the automation script:**

   ```
   Double-click: RSlip_Robot_Setup.bat
   ```

2. **Provide IP addresses** (or use defaults):
   - sbRIO IP (auto-detect or enter manually)
   - Orin IP (usually .101 or .102)

3. **Wait for 3 windows to open** → All automatic!

That's it! The script handles 23+ manual steps automatically.

📋 What Gets Automated

The main script orchestrates:

- ☑ SSH connection to sbRIO
- ☑ Starting grpccore service
- ☑ Starting fpga_driver service
- ☑ Verifying services are running
- ☑ Connecting to Jetson Orin boards
- ☑ Setting ROS2 environment variables
- ☑ Starting ROS2 bridge

📁 Files Included

File	Purpose
RSlip_Robot_Setup.bat	Main launcher - run this!
RSlip_sbRIO_Setup.bat	sbRIO SSH and service startup
RSlip_Orin1_Bridge.ps1	Jetson Orin1 ROS2 bridge setup
RSlip_Orin2_Control.ps1	Jetson Orin2 control interface
CONFIG.bat	Configuration reference (optional)

☑ Verification Checklist

After launching, verify all 3 windows:

sbRIO Window (should show):

- ☐ `grpccore` service running (one instance)
- ☐ `fpga_driver` service running (one instance)
- ☐ `TCP on 50051` confirmed

Orin1 Window (should show):

- ☐ "Orin1 Environment Setup Complete"
- ☐ CORE_MASTER_ADDR and CORE_LOCAL_IP displayed
- ☐ ROS2 bridge running without errors
- ☐ **Keep this window open!**

Orin2 Window (should show):

- ☐ Environment setup complete
- ☐ No SSH connection errors
- ☐ List of control commands displayed
- ☐ **Keep this window open!**

⚠ **If any window shows errors:** See Troubleshooting section below

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🎮 Robot Control Commands

Execute these **in Orin2 window** in order:

1. Power Up (Digital Only)

```
ros2 topic pub --once /power/command rinbo_msgs/msg/PowerCmdStamped "{header:
{seq: 1, stamp: {sec: 0, nanosec: 0}}, digital: true,signal: false,power: false}"
```

Expected: Robot powers on (lights/beeps)

2. Enable Sensors

```
ros2 topic pub --once /power/command rinbo_msgs/msg/PowerCmdStamped "{header:
{seq: 1, stamp: {sec: 0, nanosec: 0}},digital: true,signal: true,power: false}"
```

Expected: Sensors activate

3. Enable Relay (Full Power)

```
ros2 topic pub --once /power/command rinbo_msgs/msg/PowerCmdStamped "{header:
{seq: 1, stamp: {sec: 0, nanosec: 0}},digital: true,signal: true,power: true}"
```

Expected: Main power relay engages

4. Calibrate

```
ros2 run rinbo_fsm rinbo_cali
```

Expected: Feet rotate to Hall sensor position and stop. **Must complete before next step!**

5. Standing Position

```
ros2 run rinbo_fsm rinbo_standing
```

Expected: Feet rotate 180° to standing position

6. Start Tripod Gait (Experiment!)

```
ros2 run rinbo_fsm rinbo_tripod
```

Expected: Robot starts tripod motion! 🤖

🔧 Default Configuration

Component	Default Value	Notes
sbRIO IP	192.168.0.100	Customizable at startup
Orin1 IP	192.168.0.101	Auto-guessed, can change
Orin2 IP	192.168.0.102	Edit script if different
ROS2 Master Port	50051	Fixed channel
sbRIO User	admin	Fixed
sbRIO Password	admin	Fixed
Jetson User	jetson	Fixed

To change Orin2 IP permanently:

- Edit `RSlip_Orin2_Control.ps1`
- Find: `$orinIP = "192.168.0.102"`
- Change to your actual Orin2 IP

🐛 Troubleshooting

SSH Command Not Found

Error: `'ssh' is not recognized as an internal or external command`

Fix:

1. Open Settings → Apps → Apps & features
2. Find "Optional features"
3. Add "OpenSSH Client"
4. Restart PowerShell
5. Retry script

Cannot Connect to sbRIO

Symptoms: `Connection refused` or `Connection timed out`

Debugging:

```
# Check if sbRIO is online
ping 192.168.0.100

# Find sbRIO IP using ARP
arp -a | findstr "NI sbRIO"

# Or list all devices
arp -a
```

Solutions:

- Verify IP address is correct (use `arp -a` to confirm)
- Ensure sbRIO is powered on and connected to WiFi
- Check network connectivity: `ping 192.168.0.100`
- Verify SSH is enabled on sbRIO
- Try credentials manually: `ssh admin@192.168.0.100` (password: admin)

Cannot Connect to Jetson Orin

Symptoms: `Connection refused` when connecting to 192.168.0.101 or .102

Debugging:

```
# Test each Orin board
ping 192.168.0.101
ping 192.168.0.102
ping 192.168.0.103
```

Solutions:

- Verify Orin boards are powered on
- Check they're on the same WiFi network
- Get correct IP: Connect HDMI display to Jetson, login locally, run `hostname -I`
- Update IP address in script before running

Services Don't Start (grpccore/fpga_driver)

Symptoms: Services shown as "command not found" or fail immediately

Debugging:

```
# SSH into sbRIO first
ssh admin@192.168.0.100

# Check if binaries exist
ls -la /home/admin/rinbo_sbRIO_ws/install/bin/grpccore
ls -la /home/admin/rinbo_sbRIO_ws/rinbo_fpga_driver/build/fpga_driver

# Check logs
cat /tmp/grpccore.log
cat /tmp/fpga_driver.log
```

Solutions:

- Verify build process completed on sbRIO: `colcon build`
- Check if binary paths match your installation
- Review error logs for missing dependencies
- Try starting services manually to see actual error messages

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Port 50051 Not Listening

Symptoms: `NO TCP on 50051` in sbRIO window

Debugging:

```
ssh admin@192.168.0.100

# Check if grpccore is actually running
ps -ef | grep grpccore

# Try starting manually
/home/admin/rinbo_sbRIO_ws/install/bin/grpccore

# View detailed logs
tail -f /tmp/grpccore.log
```

Solutions:

- Restart sbRIO: Ensure grpccore service started properly
- Check grpccore configuration file exists
- Verify grpccore binary can run (permission issues?)
- May need to supply config path to grpccore command

Orin1/Orin2 Windows Hang or Timeout

Symptoms: Connection opens but waits indefinitely (30+ seconds)

Cause: sbRIO services not running properly

Fix:

1. Check sbRIO window first for errors
2. Ensure all services running: `ps -ef | egrep "grpccore|fpga_driver"`
3. Ensure port 50051 listening: `netstat -tn | grep 50051`
4. From the hanging Orin window, press Ctrl+C to cancel
5. Debug sbRIO issues, then retry

ROS2 Commands Not Found

Symptoms: `Command "ros2" not found` in Orin windows

Cause: ROS2 Humble not installed or environment not sourced

Fix:

```
# Verify ROS2 installed
ls /opt/ros/humble/

# Manually source in that window
source /opt/ros/humble/setup.bash
source ~/rinbo_ros_ws/install/setup.bash

# Try ros2 command again
ros2 --version
```

Robot Doesn't Respond to Power Command

Symptoms: Command executes but robot shows no response

Debugging:

```
# Verify message published
ros2 topic echo /power/command --once
```

```
# Check if message format correct
# Expected: {...digital: true, signal: false, power: false}

# If bridge not connected: Orin1 window may have disconnected
# Check Orin1 window for errors
```

Solutions:

1. Verify ROS2 bridge is connected (check Orin1 window)
2. Copy exact command from documentation (spacing/brackets matter)
3. Ensure calibration NOT required first
4. Check fpga_driver logs: `ssh admin@192.168.0.100` → `cat /tmp/fpga_driver.log`
5. Verify hardware connections/power supply

Script Freezes/Hangs

Symptoms: Something seems stuck, no output progress

Solutions:

- **Password prompt hidden:** Script may be waiting for SSH password - type it (even if not visible) and press Enter
- **Network timeout:** Press Ctrl+C to cancel, check network, retry
- **Process deadlock:** Close window, restart script

🔄 Workflow Summary

1. **Startup:** Run `RSlip_Robot_Setup.bat`
2. **Configuration:** Enter IP addresses (can auto-detect)
3. **Initialization:** Wait for 3 windows (~30 seconds)
4. **Verification:** Check all windows have no errors
5. **Power Sequence:** Run power/sensor/relay commands
6. **Calibration:** Run calibration and standing commands
7. **Execution:** Run tripod gait or custom sequence
8. **Cleanup:** Close windows after experiment

📝 Configuration Customization

Edit Network IPs

- **sbRIO IP:** Prompted at startup each time
- **Orin1 IP:** Script parameter (edit RSlip_Orin1_Bridge.ps1 if persistent change needed)
- **Orin2 IP:** Edit `RSlip_Orin2_Control.ps1`, change `$orinIP = "192.168.0.102"`

Edit Remote Paths

- Edit `RSlip_sbRIO_Setup.bat`
- Update paths if your sbRIO different setup:
  - `cd rinbo_sbRIO_ws/rinbo_fpga_driver/build/`
  - `/home/admin/rinbo_sbRIO_ws/install/bin/grpccore`

Edit Credentials

- sbRIO: Edit `RSlip_sbRIO_Setup.bat` - change `admin` username if different
- Jetson: Hardcoded as `jetson` - edit PowerShell scripts if different

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⚠ Common Issues Quick-Fix Table

Issue	Quick Fix
SSH not found	Install OpenSSH client (Win Settings)
Cannot reach sbRIO	Check IP with `arp -a`, verify power
Cannot reach Orin	Power it on, verify IP, check WiFi
Services fail to start	Check log files: `/tmp/grpccore.log`
Port 50051 not listening	Restart sbRIO services
ROS2 bridge hangs	Verify sbRIO services running first
Robot doesn't move	Check calibration done, verify connections
Command not found	Re-source ROS2: `source /opt/ros/humble/setup.bash`

🎯 Important Notes

- **All 3 windows must stay open** during entire experiment
- **Commands must run in order** (calibration before standing, etc.)
- **Calibration MUST complete** before standing position
- **Standing position MUST complete** before tripod gait begins
- Network must remain stable (WiFi shouldn't disconnect)
- SSH must be enabled on all machines
- Default credentials: sbRIO `admin/admin`, Jetson `jetson/[your-password]`

📞 Need More Help?

Quick Diagnostic:

```
# Test network connectivity from Windows
ping 192.168.0.100    # sbRIO
ping 192.168.0.101    # Orin1
ping 192.168.0.102    # Orin2
```

```
# Check ARP table for device IPs
arp -a | grep -i "192.168.0"
```

Collect for Support:

- Screenshots of error messages
- Output of all 3 windows (last 50 lines)
- Result of `arp -a` showing your devices
- Network topology (how devices connected)
- Your Windows OS version

Version: 1.0
Updated: 2026-03-31
Compatible: ROS2 Humble, sbRIO, Jetson Orin Nano
Automation: All 23+ steps condensed into one-click launch!